

Pneumonia (ICD-9 codes 480-487)

The estimated PM_{2.5} coefficient and standard error are based on a relative risk of 1.154 (95% CI -1.027, 1.298) due to a PM_{2.5} change of 36 µg/m³ in the 1-day lag GAM stringent model (Ito, 2003, Table 7).

G.1.3.4 Kloog et al. (2012)

Kloog et al. (2012) examined the relationship between both short-term and long-term PM_{2.5} exposure and emergent hospital admissions for respiratory (ICD-9 codes 460-519) and cardiovascular (ICD-9 codes 390-429) diseases in the New-England states of Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island and Vermont. The study used hospital admission data from the U.S. Medicare program for only patients that were at least 65 years old. Short-term exposure models looked at the mean 24-hour PM_{2.5} concentration on the day of a patient's hospital admission, while long-term exposure was defined to be the mean PM_{2.5} concentration over the entire study period (2000-2006). Results of the study showed an association between PM_{2.5} and hospital admissions for all disease types in both the short and long term.

Hospital Admissions, Respiratory (ICD Codes 460-519)

In a single pollutant model for patients over the age of 65, the coefficient and standard error were estimated from the percent change (0.70%) and 95% confidence interval (0.35%-1.05%) for a 10 µg/m³ increase in same day (0 lag) 24-hour mean PM_{2.5} concentration.

G.1.3.5 Moolgavkar (2000a)

Moolgavkar (2000a) examined the association between air pollution and COPD hospital admissions (ICD 490-496) in the Chicago, Los Angeles, and Phoenix metropolitan areas. He collected daily air pollution data for ozone, SO₂, NO₂, CO, and PM₁₀ in all three areas. PM_{2.5} data was available only in Los Angeles. The data were analyzed using a Poisson regression model with generalized additive models to adjust for temporal trends. Separate models were run for 0- to 5-day lags in each location. Among the 65+ age group in Chicago and Phoenix, weak associations were observed between the gaseous pollutants and admissions. No consistent associations were observed for PM₁₀. In Los Angeles, marginally significant associations were observed for PM_{2.5}, which were generally lower than for the gases. In co-pollutant models with CO, the PM_{2.5} effect was reduced. Similar results were observed in the 0-19 and 20-64 year old age groups.

The PM_{2.5} C-R functions are based on the single and co-pollutant models (PM_{2.5} and CO) reported for the 20-64 and 65+ age groups. Since the true PM effect is most likely best represented by a distributed lag model, then any single lag model should underestimate the total PM effect. As a result, we selected the lag models with the greatest effect estimates for use in the C-R functions.

Hospital Admissions, Chronic Lung Disease Less Asthma (ICD-9 codes 490-492, 494-496)

In a model with CO, the coefficient and standard error are calculated from an estimated percent change of 2.0 and t-statistic of 2.2 for a 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} in the two-day lag model (Moolgavkar, 2000a, Table 4, p. 81). In a log-linear model, the percent change is equal to $(\text{RR} - 1) * 100$.

In this study, Moolgavkar defines and reports the “estimated” percent change as $(\log \text{RR} * 100)$. Because the relative risk is close to 1, $\text{RR}-1$ and $\log \text{RR}$ are essentially the same. For example, a true percent change of 2.0 would result in a relative risk of 1.020 and coefficient of 0.001980. The “estimated” percent change, as reported by Moolgavkar, of 2.0 results in a relative risk of 1.020201 and coefficient of 0.002.

Note that although Moolgavkar (2000a) reports results for the 20-64 year old age range, for comparability to other studies, we apply the results to the population of ages 18 to 64. Note also that in order to avoid double counting non-elderly asthma hospitalizations (ICD code 493), which are typically estimated separately in EPA benefit analyses, we have excluded ICD code 493 from the baseline incidence rate used in this function.

G.1.3.6 Moolgavkar (2000b), Cardiovascular

Moolgavkar (2000b) examined the association between air pollution and cardiovascular hospital admissions (ICD 390-429) in the Chicago, Los Angeles, and Phoenix metropolitan areas. He collected daily air pollution data for ozone, SO₂, NO₂, CO, and PM₁₀ in all three areas. PM_{2.5} data was available only in Los Angeles. The data were analyzed using a Poisson regression model with generalized additive models to adjust for temporal trends. Separate models were run for 0 to 5 day lags in each location. Among the 65+ age group, the gaseous pollutants generally exhibited stronger effects than PM₁₀ or PM_{2.5}. The strongest overall effects were observed for SO₂ and CO. In a single pollutant model, PM_{2.5} was statistically significant for lag 0 and lag 1. In co-pollutant models with CO, the PM_{2.5} effect dropped out and CO remained significant. For ages 20-64, SO₂ and CO exhibited the strongest effect and any PM_{2.5} effect dropped out in co-pollutant models with CO.

Hospital Admissions, All Cardiovascular (ICD codes 390-409, 411-429)

The single pollutant coefficient and standard error are calculated from an estimated percent change of 1.4 and t-statistic of 4.1 for a 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} in the zero lag model (Moolgavkar, 2000b, Table 4, p. 1203).

Note that (Moolgavkar (2000b) report results that include ICD code 410 (heart attack). In a benefit analysis, avoided nonfatal heart attacks are typically estimated separately. The baseline rate in the Peters et al. function is a modified heart attack hospitalization rate (ICD code 410), since most, if not all, nonfatal heart attacks will require

hospitalization. In order to avoid double counting heart attack hospitalizations, we have excluded ICD code 410 from the baseline incidence rate used in this function.

G.1.3.7 Moolgavkar (2003)

Moolgavkar (2000a) examined the association between air pollution and COPD hospital admissions (ICD 490-496) in the Chicago, Los Angeles, and Phoenix metropolitan areas. In response to concerns with Splus issue, Moolgavkar (2003) reanalyzed his earlier studies. In the reanalysis, he reported that more generalized additive models with stringent convergence criteria and generalized linear models resulted in smaller relative risk estimates.

Hospital Admissions, Chronic Lung (ICD-9 codes 490-496)

The coefficient and standard error are calculated from an estimated percentage change of 1.85 and a t-statistic of 3.53 for a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ in the 2-day lag GAM-30df stringent (10-8) model (Moolgavkar, 2003, Table 17). In a log-linear model, the percent change is equal to $(\text{RR} - 1) * 100$.

The $\text{PM}_{2.5}$ C-R functions for the 65+ age group are based on the reanalysis in Moolgavkar (2003) of the single and co-pollutant models ($\text{PM}_{2.5}$ and CO). The true PM effect is most likely best represented by a distributed lag model, then any single lag model should underestimate the total PM effect. As a result, we selected the lag models with the greatest effect estimates for use in the C-R functions.

Hospital Admissions, All Cardiovascular (ICD-9 codes 390-429)

The single pollutant coefficient and standard error are calculated from an estimated percent change of 1.58 and t-statistic of 4.59 for a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ in the 0-day lag GAM-30df stringent (10-8) model (Moolgavkar, 2003, Table 12). In a log-linear model, the percent change is equal to $(\text{RR} - 1) * 100$.

G.1.3.8 Peng et al. (2008)

Peng et al. (2008) examined the risk of hospital admissions for cardiovascular and respiratory diseases in relation to particulate matter ($\text{PM}_{10-2.5}$ and $\text{PM}_{2.5}$). To accomplish this, the authors utilized a database of 108 U.S. counties with daily emergency hospital admission rates for cardiovascular and respiratory diseases among Medicare enrollees living 9 miles from air monitors, temperature, and dew-point temperature. $\text{PM}_{10-2.5}$ and $\text{PM}_{2.5}$ concentrations were calculated by using monitoring data from January 1, 1999 through December 31, 2005. Overall, there were 3.7 million cardiovascular disease and 1.4 million respiratory disease-related hospital admissions for the time period assessed. The authors found that a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{10-2.5}$ was associated with a 0.36% increase (95% PI: 0.05-0.68%) in cardiovascular disease admissions on the same day, and a 0.25% increase (95% PI: -0.11-0.60%) after adjusting for $\text{PM}_{2.5}$. For respiratory disease admissions, a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{10-2.5}$ was found to be associated with an unadjusted 0.33% increase in respiratory disease admissions (95% PI: -0.21- 0.86%) and an adjusted 0.26% increase (95% PI: -0.32-

0.84%) in emergency admissions. Also, unadjusted associations of PM_{2.5} with cardiovascular and respiratory disease admissions were 0.71% (95% PI: 0.45-0.96%) for same-day exposure and 0.44% (95% PI: 0.06-0.82%) for exposure lagged by 2 days prior to hospital admission.

Hospital Admissions, Cardio-, Cerebro-, and Peripheral Vascular Disease (ICD-9 codes 426-427, 428, 430-438, 410-414, 429, 440-448)

In a single-pollutant model, the coefficient and standard error are estimated from the percent change in daily admission (0.44%) and 95% posterior interval (95% PI: 0.06-0.82%) for a 10 µg/m³ increase in daily 24-hour mean PM_{2.5} concentrations for the same day (Peng et al., 2008, page 2175).

Note that Peng et al. (2008) considered a broader range of ICD-9 codes and estimated the risk of both cardiovascular events and cerebro- and peripheral vascular disease. For comparability to other studies, EPA decided to apply a baseline hospitalization rate for ICD-9 codes 390-409 and 411-429 when using this C-R function in quantifying impacts.

G.1.3.9 Peng et al. (2009)

Peng et al. (2009) investigated the relationship between hospital admissions for cardiovascular and respiratory disease and the chemical components of PM_{2.5} across 119 U.S. urban communities for 12 million Medicare enrollees using log-linear Poisson regression models. This was achieved using a national database with daily data from 2000-2006 on emergency hospital admissions of cardiovascular and respiratory outcomes, ambient levels of PM_{2.5} components and weather variables. Bayesian hierarchical statistical models were used to estimate the associations. Three scenarios for PM_{2.5} exposure were assessed which were as follows: 1) for the period 2000-2006 and including only days with available measurements for all 7 PM_{2.5} components from the Speciation Trends network (STN); 2) PM_{2.5} measured by the STN for the period 2000-2006 and including only days with available measurements for all 7 PM_{2.5} components from the STN and 3) PM_{2.5} estimated as the sum of the 7 largest components of PM_{2.5} mass for the period 2000-2006. Results of percent increases in emergency admissions associated with PM_{2.5} at lag 0 under these scenarios were showed in Figure 2 and the results for the components of PM_{2.5} from both single and multi-pollutant models were showed in Figure 3. In multi-pollutant models that adjusted for the levels of other pollutants, the authors found that an interquartile range increase in elemental carbon was associated with a 0.80% increase (95% PI: 0.34-1.27%) in risk of same-day cardiovascular admissions. Similarly, an interquartile range increase in organic carbon matter was associated with a 1.01% increase (95% PI: 0.04-1.98%) risk of respiratory admissions on the same day.

Hospital Admissions, Cardio-, Cerebro-, and Peripheral Vascular Disease (ICD-9 codes 426-427, 428, 430-438, 410-414, 429, 440-448)

In a single-pollutant model, the coefficient and standard error are estimated from the percent change in daily admission (0.68%) and 95% posterior interval (95% PI: 0.26-